Prototype and priority pathogens: how the frameworks differ

Reference note prepared 20 October 2026, revised after a completeness and fact-check pass. Every substantive claim carries a link to its primary source; the handful that do not are marked ***[unsourced]*** *or* ***[not retrieved]*** *in place. Where the underlying documents disagree with each other, or where a claim could not be verified, that is stated rather than smoothed over. Scope: viral prioritisation frameworks, with bacterial and fungal lists named where a framework treats them as peers but not analysed in the matrix.*

1. The conceptual distinction

A **priority pathogen** is selected by risk. The question it answers is: *how bad would this be, and how likely is it?* The countermeasure developed against it is intended to be used against it. A **prototype pathogen** is selected by representativeness and tractability. The question it answers is: *if we build tools against this one, do they transfer to its relatives — including relatives we have not met — and can we actually work on it today?* The countermeasure developed against it may never be deployed against it; the real deliverable is a transferable method.

These are not two tiers of the same ranking. They are orthogonal criteria, and the bodies that use both say so explicitly. NIAID, which originated the modern usage, states that the two designations "do not necessarily overlap" and gives a worked example in each direction in its Pandemic Preparedness Plan: Ebola virus is both, whereas "Zika virus of the Flaviviridae family is known to cause human morbidity, but it does not serve as a good prototype for development of MCMs for other Flaviviridae family members. Thus, NIAID will continue to support Zika virus research in the capacity of a priority pathogen."

WHO adopted the vocabulary in 2024 and gave it formal definitions in Figure 3 of *Pathogens prioritization: a scientific framework for epidemic and pandemic research preparedness*. A Priority Pathogen is one where "existing knowledge of patterns of transmission, virulence, and access to MCMs suggests a Pathogen can cause a PHEIC (or a pandemic)". A Prototype Pathogen is a "representative pathogen within a family selected to serve as a model for fundamental research & translational research to develop MCM[s] that can be applied to other members of the family". (The figure as printed contains two typographical errors — "MCMMs" and "couls" — so it should not be quoted as verbatim without `[sic]`.)

The mismatch runs in both directions and the examples are concrete.

Prototype but not priority. WHO names *Orthopoxvirus vaccinia* as a Poxviridae prototype alongside monkeypox, on the published ground that "vaccinia provides BSL2 alternative that is cross-protective and likely susceptible to same antivirals". It names the genus *Vesiculovirus* as the Rhabdoviridae prototype because it is "important as vaccine vector and as potential prototype for future vaccines" — a tool-value criterion, not a threat criterion, applied to a family WHO rates Low risk and for which it proposes no priority pathogen at all. West Nile virus and tick-borne encephalitis virus are Flaviviridae prototypes but not priority pathogens, selected to span Culex and tick transmission respectively. NIAID's list contains Hazara virus (a BSL-2 surrogate for CCHFV with an existing reverse-genetics system), Cedar virus (a non-pathogenic henipavirus), canine distemper virus (an animal morbillivirus) and Punta Toro virus. None is a public-health threat.

Most striking of all: Rift Valley fever virus was a WHO *priority disease* in 2015, 2017 and 2018 and is a *prototype only* in 2024 — a demotion on the risk axis with retention on the research axis. **But the demotion is WHO's alone.** CEPI's live priority pathogens page still names Rift Valley fever as a standing priority pathogen, and the CEPI 3.0 Investment Case (Annex 2, p.41) carries it as a current priority pathogen under "Non-Arenavirus Bunyaviricetes" alongside CCHF and SFTS, with a commitment to "Complete Phase 2 clinical trials and establish outbreak-ready investigational reserves for high-risk pathogens including Nipah, MERS and Rift Valley fever" (p.6). The same organism is simultaneously a WHO prototype-only designation and a live CEPI procurement-stage priority. That is the showpiece example in full: it is not a demotion, it is a divergence.

Priority but not prototype. WHO lists *Orthomarburgvirus marburgense* and *Orthoebolavirus sudanense* as priority pathogens while naming only *Orthoebolavirus zairense* as the Filoviridae prototype. Yellow fever virus is a priority pathogen with no prototype role. Variola is a priority pathogen and explicitly not a prototype; WHO's stated reason sits in the monkeypox prototype note — "Orthopox monkeypox needed to study pathogenesis. Impossible to work with variola except in special settings" — i.e. a tractability bar excluded the more dangerous organism. Five of WHO's seven priority influenza A subtypes are not prototypes.

And the two frameworks disagree about the same organism. NIAID says Zika is a priority pathogen but a poor Flaviviridae prototype. WHO designates *Orthoflavivirus zikaense* as both a priority pathogen and a prototype. Same virus, same concept, opposite call.

The practical consequence for anyone reading these documents: a prototype list is not a ranking of danger and must never be cited as one. NIAID's prototype list in particular excludes the two highest-risk viral families on earth *because* they are well resourced. A reader who treats that list as NIAID's threat assessment will conclude NIAID does not rate coronaviruses or influenza, which is the opposite of what the documents say.

2. Who publishes what, and what for

ORGANISATION	FRAMEWORK	UNIT OF PRIORITISATION	PURPOSE	LAST REVISED	USES "PROTOTYPE"?
WHO (R&D Blueprint)	Pathogens prioritization: a scientific framework	Viral family, with Priority Pathogen and Prototype Pathogen nested inside; named unit is often subgenus, genus, genotype or HA-subtype rather than species	R&D steering and research-agenda coordination	Cover date June 2024; published 30 July 2024; PDF silently re-issued July 2025 (internal title "V7") with prioritisation tables unchanged	Yes — defines it
WHO (AMR)	Bacterial Priority Pathogens List 2024 and Fungal Priority Pathogens List 2022	Pathogen–drug combination (BPPL); species in Critical/High/Medium groups (FPPL)	R&D steering and public-health action on AMR	BPPL May 2024; FPPL Oct 2022	No — WHO declined to extend prototypes to bacteria
NIAID	Pandemic Preparedness Plan + Deschamps et al. 2023, *J Infect Dis* 228(Suppl 6):S355–S358	Prototype-within-family (10 families) and priority pathogen at species level	R&D steering and funding allocation (ReVAMPP, AViDD)	Plan Dec 2021 (page last reviewed 15 Dec 2021); prototype list Oct 2023	Yes — originated it
NIAID (biodefense)	Category A / B / C Priority Pathogens, at `niaid.nih.gov/research/emerging-infectious-diseases-pathogens`	Named agent, in three tiers derived from the CDC bioterrorism categories	Biodefense research prioritisation	**[not retrieved]** — the page returned HTTP 403 on 20 Oct 2026; it is cited as live by both HERA (Figure 1 column "US NIH NIAID's Biodefense Pathogens") and by Pulik et al. ("the US National Institute for Allergy and Infectious Diseases (NIAID) published a biodefence list")	No
UKHSA	Priority pathogen families R&D tool	Pathogen family (16 viral + 8 bacterial), with one to four illustrative "notable pathogens" each	R&D steering for England-based funders under the UK Biological Security Strategy	Published 25 Mar 2025; last updated 8 Apr 2025 (the GOV.UK page states the reason: a page-11 edit reflecting the change in HCID status of clade I mpox)	No — explicitly declines
CEPI	Priority pathogens + CEPI 3.0 Strategy 2027–31 + CEPI 3.0 Investment Case	Mixed: individual pathogen (current), viral family (from 2027), prototype-within-family (Vaccine Library)	Vaccine-investment portfolio, gated on market failure as well as risk	CEPI 3.0 launched 13 Feb 2026, covering 2027–31	Yes — in three incompatible senses
European Commission / HERA	SWD(2026) 1 final	Viral family, tiered (5 highest / 7 high); AMR by pathogen–drug combination; CBRN by agent category	MCM intervention across the whole lifecycle — R&D **through procurement and stockpiling**	8 Jan 2026; content current to 5 Oct 2025; next update "during 2027"	Yes — imports WHO's designations
ECDC	EU epidemiological surveillance under Commission Implementing Decision (EU) 2018/945 + the ECDC tool for the prioritisation of infectious disease threats (MCDA handbook, Stockholm 2017)	Notifiable communicable disease (2018/945); whatever the user defines (the tool is a method, not a list)	Surveillance, and a transferable prioritisation instrument for member states	2018/945 as amended; handbook Aug 2017	No
US HHS/ASPR (PHEMCE)	High-Priority Threats + Multiyear Budget FY2025–29	Named agent (legacy deliberate-release threats) **and** viral family (the emerging-infectious-disease bucket)	**Procurement requirements-setting** plus R&D steering; gated by DHS Material Threat Determinations	SIP Sept 2024; MYB July 2026	Only as a description of what NIH does
US CDC/USDA	Select Agents and Toxins List, as amended by 89 FR 101941	Regulated biological material, at whatever rank the regulation needs (species, strain, subtype, construct)	**Biosafety and biosecurity regulation** — who may possess, use and transfer	Rules effective 16 Jan 2025	No

ORGANISATION	FRAMEWORK	UNIT OF PRIORITISATION	PURPOSE	LAST REVISED	USES "PROTOTYPE"?
Japan MHLW	重点感染症リスト	Pathogen group within a *foreseeability* tier (Groups X/A/B/C/D)	R&D steering and procurement, gating SCARDA/AMED funding and expedited PMDA review	26 March 2025	No — but uses the logic
Africa CDC	Continental ranking plus regional rankings. Three are published with retrievable ranked lists (Eastern, Northern, Southern). Exercises for the Western and Central regions were run in Abidjan (24–26 June 2025) and Yaoundé (8–10 July 2025), but ECDC's report of 15 Sept 2025 records only that participants "reached consensus on a set of priority diseases" and publishes no ranked list	Disease	Emergency preparedness and resource allocation	Continental page dated 22 Apr 2026	No
Academic dissent	Adalja & Inglesby, *OFID* 2025	Viral family (six, unranked)	Argument that the other lists are too broad	26 May 2025	Discusses it critically

The single biggest source of apparent disagreement is purpose, and specifically whether a list serves procurement or research. Three of these are not R&D lists at all. The Select Agents list is a biosafety regulation under 42 USC 262a: it tells you what you may keep in a freezer, not what the government is funding. Its current form was set by the HHS/CDC final rule at 89 FR 101941 (17 Dec 2024), effective 16 January 2025, which lists at §73.3(b)(19) "Severe acute respiratory syndrome coronavirus (SARS-CoV)" and at (20) "SARS-CoV/SARS-CoV-2 chimeric viruses resulting from any deliberate manipulation of SARS-CoV-2 to incorporate nucleic acids coding for SARS-CoV virulence factors" — but not SARS-CoV-2 itself and not MERS-CoV. The only human influenza entry, at (16), is "Reconstructed replication competent forms of the 1918 pandemic influenza A virus containing any portion of the coding regions of all eight gene segments". PHEMCE's list is gated by DHS Material Threat Determinations, which are an appropriations key: Project BioShield money can only be spent against MTD threats, which is why pandemic influenza — the largest single funded portfolio at $15.65bn over five years (MYB p.20 and Appendix C p.41) — carries no asterisk on the published list and is funded through a separate dedicated appropriation instead. HERA sits in between: its stated objective is "to identify the priority threats that require interventions to address vulnerabilities and strategic dependencies related to the development, production, procurement, stockpiling and distribution of MCM", and it folds existing MCM availability into the prioritisation itself rather than reporting it as an output.

Within the R&D lists, each does something the others do not.

WHO is the only body that rates the *family itself* as an object (Low / Low-to-Medium / Medium / High PHEIC risk), including families with no priority pathogen at all, and the only one that publishes a per-region breakdown across all six WHO regions. Its stated rationale for prioritising below the resource-rich families is the "streetlight effect": disease-by-disease lists concentrate effort on "well-studied pathogens with readily accessible data" and neglect the "Dark Areas". It also pairs the list with a delivery mechanism — on the 7 April 2026 roadmap launch page, whose table is headed "Current list of CORCs (as of March 2026)", **seven viral Collaborative Open Research Consortia covering ten viral families, plus one bacterial consortium**. The viral seven are Arenaviridae (UKHSA), Bunyavirales (UKHSA with SCARDA and Institut Pasteur de Dakar, covering Hantaviridae, Nairoviridae, Peribunyaviridae and Phenuiviridae — four families in one consortium, which is how seven consortia cover ten families), Coronaviridae (PREPARE Singapore), Filoviridae (ANRS-MIE France), Flaviviridae (Fiocruz Brazil), Paramyxoviridae (ICMR India) and Poxviridae (INRB DR Congo with Wits RHI South Africa). The bacterial consortium is hosted by Aga Khan University and LSHTM. WHO also publishes separate bacterial and fungal priority pathogen lists, which HERA treats as peer frameworks in its Chapter III (sections 3.1 and 3.2) and which Pulik et al. count among their thirteen resources; neither uses the prototype concept.

NIAID is the only framework that deselects families *because* they are well resourced. Its stage-one criteria, set out in Deschamps et al. (2023), were "(1) potential to cause pandemics, (2) the status of MCMs, and (3) resources devoted to their characterization", with (2) and (3) as inverse criteria, combined in a two-by-two of pandemic potential against existing resources from which the lower-right quadrant was selected (that article's Figure 1, adapted from Cassetti et al.). It is also the only one that devolves final prototype choice to funded investigators: RFA-AI-23-020 requires applicants to justify their own prototype against published considerations rather than fixing it centrally. Note two terminological traps. First, NIAID uses "priority" at *two* levels — "Priority Viral Families for Pandemic Preparedness" is the header on the ten-family prototype table, while "priority pathogen" denotes the separate species-level track. Second, NIAID maintains **three** parallel lists, not two: the prototype-family table, the species-level priority pathogens, and the Category A/B/C biodefense list, which is the one HERA's Figure 1 cross-checks against under the column head "US NIH NIAID's Biodefense Pathogens". The biodefense list could not be retrieved for this note (see table above).

UKHSA is the only framework that deliberately mixes imported threats with England-endemic disease, and the only one that embeds a countermeasure-availability audit at assay-format granularity (PCR/serology/LFD, throughput, point-of-care, research-use-only). It is also the only body that names the prototype concept and abstains: "Some of these pathogens have been suggested elsewhere in the scientific literature as so-called 'prototype pathogens' or their use advocated as pathfinders... UKHSA has not done its own assessment of the use of these pathogens for this purpose."

CEPI is an investment portfolio, not a research agenda, and is gated on market failure. This is why influenza — which CEPI itself calls "the only disease that we can say with certainty will cause a future pandemic" (Investment Case, p.42) — is not on its list.

A note on the status of CEPI 3.0 claims in this document. CEPI 3.0 is a forward strategy covering 2027–31. Its launch on 13 February 2026 is confirmed by a live CEPI press release and a live landing page, and the substantive claims below are drawn from two separate published documents — the Strategy report and the Investment Case — rather than from one PDF. The financial figures quoted in sections 5 and 6 (US$214.5m across 14 broadly protective coronavirus vaccine projects; US$54.3m to Moderna's mRNA H5 Phase 3) come from a third document, CEPI's 2025 Annual Progress Report, and so do not depend on the CEPI 3.0 documents at all. What remains genuinely open is that CEPI's live priority pathogens page has not been updated to reflect any of this: it names eight pathogens plus Disease X, its strategy navigation still reads "CEPI 2.0 and the 100 Days Mission", and it does not mention CEPI 3.0. **Readers should treat the family-level CEPI entries in section 3 as the 2027–31 intention stated in the Investment Case, not as the portfolio CEPI's own public-facing list currently describes.** The two are not consistent, and CEPI has not reconciled them.

HERA is a meta-list. It starts from binding EU legal instruments — Commission Implementing Decision (EU) 2018/945, the epidemiological-surveillance list ECDC operates, and the dual-use Delegated Regulation (EU) 2022/1 — filters to risk groups 3 and 4 under Directive 2000/54/EC, and then cross-checks against WHO (2024 and 2018), US HHS Select Agents, NIAID's Biodefense Pathogens, NIAID's Antiviral Program for Pandemics and CEPI. Its Figure 1 has thirteen cross-check columns, of which five carry WHO designations: "viral families with high 'PHEIC risk'", "priority pathogen (global)", "prototype pathogen (global)", "selected priority pathogens with circulation in WHO EURO region", "selected prototype pathogens with circulation in WHO EURO region" — and a sixth, dated 2018, headed "R&D blueprint 'priority pathogens'". So Figure 1 does separate priority from prototype into distinct columns, and does so twice over (globally and for EURO), and it carries its own temporal comparison.

Two cautions, both of which the draft of this note previously got wrong in opposite directions. First, the figure *additionally* applies a merged bold convention in its agent-name column, and the legend describes only that convention: "Boldened agents are specifically listed by the WHO either as priority or prototype pathogens." Anyone reading the legend alone will conclude the two designations are merged; anyone reading the column heads will see they are not. Both are true of different parts of the same figure, and a reader quoting the legend as a description of the whole figure will be wrong. Second, the claim that this is the *only* published cross-walk separating the two is a negative claim about the whole literature and cannot be established by a survey of this size. The accurate statement is: no other cross-walk separating WHO's priority and prototype designations was found in this survey, and HERA's is the only one found — which is a bounded claim, and the one section 7 now makes.

Crucially, HERA makes the prototype concept load-bearing in a procurement instrument: "available" MCM status is defined as a marketing authorisation "with an approved indication for use against the prototype pathogen(s) of that viral family".

3. The overlap matrix

Presence and unit, by family. WHO 2024 entries give the family's PHEIC risk rating and the unit at which priority (P) and prototype (X) entries are named. The **WHO 2018** column records whether the family had a representative on the legacy R&D Blueprint priority-diseases list, taken from HERA's Figure 1 column "R&D blueprint 'priority pathogens' 2018" read against the 2018 annual review.

Two columns compare units that are not families and are marked accordingly. **Africa CDC** and **Japan MHLW** both prioritise at disease level; their cells record whether a disease of that family appears, which is a weaker statement than the family-level columns make. This is not a presentational quibble — see the unit-of-

analysis discussion in 3(b).

FAMILY	WHO 2024	WHO 2018	NIAID PROTOTYPE PROGRAMME	UKHSA (PANDEMIC / EPIDEMIC)	CEPI (2027–31 INTENT)	HERA	PHEMCE	AFRICA CDC (DISEASE LEVEL)	JAPAN MHLW (DISEASE LEVEL)	ADALJA & INGLESBY
Coronaviridae	High. P + X = **subgenera** Sarbecovirus, Merbecovirus	Yes — SARS, MERS	**Excluded by name**	High / High — MERS-CoV	Yes; priority MERS-CoV; pan-sarbecovirus track	Group 1	**EID family (2nd listed)**	Yes — rank varies by region	Groups A and B	Yes
Orthomyxoviridae	High. P = 7 **HA subtypes**; X = H1N1, H5Nx	**No — explicitly out of scope**	**Excluded by name**	High / High — "non-seasonal influenza"	**Named gap**; flagged a "strong candidate for inclusion"	Group 1	**EID family (1st listed)**, plus a standalone threat with no MTD	Yes — top five in all three published regions	Groups A and B	Yes
Paramyxoviridae	High. P + X = Nipah	Yes — Nipah	Yes — 4 prototypes	High / High — Nipah	Yes; priority Nipah	Group 2	**EID family**	Not in published top five	Group B (Nipah)	Yes
Picornaviridae	**Medium**. P = poliovirus; X = EV-A71, EV-D68	No	Yes — 4 prototypes	High / High — EV-D68/A71	**Named gap**	Group 2 (only non-High-risk entry)	**Not listed**	No	No	Yes
Pneumoviridae	Low–Medium. X = *Metapneumovirus hominis*	No	No	Medium / High — hMPV	No	**Explicitly not prioritised**	Not listed	No	Group B (RSV)	Yes
Adenoviridae	Low–Medium. X = 2, incl. a functional category	No	No	Medium / High	No	Not listed	Not listed	No	No	Yes
Filoviridae	High. P = 3; X = Ebola Zaire	Yes — Ebola, Marburg	Yes — but absent from ReVAMPP's funded scope	Low / High — EBOV, SUDV, Marburg	Yes; priority Ebola Zaire, Marburg, plus broadly protective filovirus vaccines	Group 1	**EID family**, *and* Ebolavirus and Marburgvirus separately as MTD-asterisked named agents	Yes — Ebola and Marburg rank 1st–3rd in Eastern Africa	Group B	No
Poxviridae	High. P = variola, mpox; X = mpox, **vaccinia**	Yes — smallpox (as out of scope)	No	Medium / High — clade I MPXV	Yes; priority mpox	Group 1	**EID family**, *and* Variola separately as an MTD-asterisked named agent	Mpox ranks high continentally	Group A (smallpox), Group B (mpox)	No
Flaviviridae	High. P = 3; X = 4	Yes — but dengue, yellow fever and West Nile all declared out of scope	Yes — 3 prototypes	Low / High	**Named gap**; flagged a "strong candidate for inclusion"	Group 1	**EID family**	Yellow fever and dengue appear regionally	No	No
Arenaviridae	High. P = Lassa; X = 3	Yes — Lassa	Yes — Lassa, Junín	Low / Medium — Lassa	Yes; priority Lassa, Junín	Group 2	**EID family**	Lassa ranks in Western Africa (list unpublished)	Group B (Lassa)	No
Togaviridae	High. P + X = chikungunya, VEEV	No	Yes — 2 prototypes	Low / Medium	Yes; priority chikungunya	Group 2	**EID family**	Chikungunya appears regionally	No	No

FAMILY	WHO 2024	WHO 2018	NIAID PROTOTYPE PROGRAMME	UKHSA (PANDEMIC / EPIDEMIC)	CEPI (2027–31 INTENT)	HERA	PHEMCE	AFRICA CDC (DISEASE LEVEL)	JAPAN MHLW (DISEASE LEVEL)	ADALJA & INGLESBY
Phenuiviridae	High. P = SFTSV; X = SFTSV, RVFV	Yes — RVF	Yes — 3 prototypes	Low / Medium	**Yes — priority RVF *and* SFTS** (Investment Case Annex 2)	Group 2	Not listed	RVF appears regionally	Group B (SFTS)	No
Nairoviridae	High. P + X = CCHFV	Yes — CCHF	Yes — CCHFV, **Hazara**	Low / Medium — CCHFV	Yes; priority CCHF	Group 2	Not listed	CCHF appears regionally	Group B (CCHF)	No
Hantaviridae	High. P = 2; X = Sin Nombre	No	Yes — 3 prototypes	**Low / Low**	Named in the Investment Case prose list of nine families but **absent from its Annex 2 table**; no named pathogen, no identified project	Group 2	Not listed	No	No	No
Peribunyaviridae	**Low**. No P; X = Oropouche	No	Yes — 2 prototypes	Low / Medium — Oropouche	No	"Under consideration"	Not listed	No	No	No
Rhabdoviridae	**Low. No priority pathogen proposed**; X = genus *Vesiculovirus*, present in all six WHO regions	No	No	**Not listed**	No	**Explicitly not prioritised** (rabies)	Not listed	**Rabies is the joint most frequently prioritised pathogen worldwide — 10 of 13 resources in Pulik et al.**	**Group D** (imported, alongside malaria)	No
Caliciviridae	**Not among the families assessed**	No	No	Medium / High — norovirus	No	**Explicitly not prioritised**	Not listed	No	No	No
Retroviridae	Medium. P + X = HIV-1	Yes — HIV/AIDS (as out of scope)	No	No	No	**Explicitly not prioritised**	Not listed	No	No	No

HERA's full set of explicitly non-prioritised families. The Figure 1 note names seven, not three: "Retroviridae (HIV), Hepadnaviridae (Hepatitis B), Rhabdoviridae (rabies), Caliciviridae (norovirus), Reoviridae (rotavirus), Pneumoviridae (RSV) and Herpesviridae (varicella)." Hepadnaviridae, Reoviridae and Herpesviridae are not given matrix rows here because no other framework in the matrix carries them either, but they belong in any count of what HERA rejected.

(a) The consensus core

Three families appear in every general framework surveyed: **Coronaviridae, Orthomyxoviridae and Paramyxoviridae**. This is a change from the four-family core this note previously claimed, and the reason is the addition of PHEMCE as a column: Picornaviridae has no entry in the "#" footnote on ASPR's Priority Threats page, which lists Orthomyxoviridae, Coronaviridae, Paramyxoviridae, Poxviridae, Flaviviridae, Filoviridae, Togaviridae, Arenaviridae and Antimicrobial Resistant Pathogens. Picornaviridae is therefore absent from one of the eleven frameworks and is not in the core.

It would be possible to recover the four-family core by defining "general framework" to exclude procurement bodies. This note does not do that, because the definition would be self-serving: section 2's central thesis is that procurement-versus-research is the main axis of disagreement, and a core defined by excluding the procurement bodies is a core that has assumed away the thing under study. The honest statement is that the core is three if PHEMCE counts and four if it does not, and that the one family whose membership turns on the definition is Picornaviridae.

Note also that the three-family core is still exactly three of the four families UKHSA rates High for *both* pandemic and epidemic potential, and three of Adalja & Inglesby's six.

The empirical record agrees at the pathogen level. Pulik et al., *Scoping review on the prioritisation of high-consequence infectious pathogens for research preparedness and response to health emergencies*, **BMC Medicine** 2026;24:301, published 1 April 2026 reviewed 13 prioritisation resources published since 2018 and found "Rabies and influenza were the most often listed viruses and pathogens overall, appearing in 10 resources each". Thirty-eight different viral haemorrhagic fevers and arboviruses were featured across 10 resources, with Ebola and Marburg most often named individually (six and five mentions respectively) — note that this is a count of *resources mentioning any member of a large heterogeneous group*, not a 10/13 ranking for the group as a unit, which is how this note previously reported it. Coronaviruses fragment: SARS appears in 5/13, MERS in 5/13, SARS-CoV-2 in only 3/13 — partly because two of the three resources naming SARS and MERS predate 2019, and partly because several national zoonotic exercises treat SARS-CoV-2 as established rather than emerging.

Three caveats on that apparent consensus, all from the review itself. First, **7 of Pulik's 13 resources were produced using a single shared instrument**, the US CDC One Health Zoonotic Disease Prioritization process; convergence between national lists is therefore substantially an artefact of shared methodology rather than independent agreement. Second, the included set is not the same set this note compares: it contains the NIAID biodefence list and a WHO AMR list, but not UKHSA, CEPI, HERA or ECDC. Third, the authors record their own geographical blind spot: the Overton search "identified resources from Latin America and South-East Asia, none met the illegibility criteria as they did not contain lists of priority pathogens, so these areas are not represented in our analysis", and they name a South-East Asian regional exercise (the ASEAN strategy) that their search missed because "Overton do not scrap the Association of South-East Asian Nations website". The regional-divergence argument in 3(b) below therefore runs on Africa CDC alone and inherits that blind spot, which is now stated rather than inherited silently.

(b) The genuine disagreements, and what drives them

Resourcing, applied as an inverse criterion. The sharpest divergence in the whole landscape is NIAID's deliberate exclusion of Coronaviridae and Orthomyxoviridae from its prototype programme. The published sentence, from Deschamps et al. (2023), *J Infect Dis* 228(Suppl 6):S355–S358, doi:10.1093/infdis/jiac465, reads: "Although Orthomyxoviridae and Coronaviridae have a very high pandemic risk, they were not included in this prototype pathogen exercise since significant resources are already allocated to characterize these families, and there are licensed or advanced clinical stage development MCMs available for viruses within them." Cassetti et al. (2022) put it differently: they are excluded "because they are considered priority pathogens". Either way, the exclusion is a statement about the *research* portfolio, not about threat, and it is the mirror image of WHO's position.

Mandate and market failure. CEPI excludes Orthomyxoviridae, Flaviviridae and Picornaviridae while rating influenza at or above everything on its list, because its gate is market failure. Its own words, from the Investment Case p.41: "several important families – such as Orthomyxoviridae, Flaviviridae and Picornaviridae – remain outside CEPI's current scope, representing ongoing strategic gaps that a fully funded CEPI 3.0 strategy may be poised to fill." On influenza specifically (p.42): "Influenza is widely regarded as the most likely cause of the next pandemic, and we agree with the scientific consensus that flu is one of the most important threats we face and the only disease that we can say with certainty will cause a future pandemic. Because of this, there has long been significant investment in this space." CEPI's Annex 2 adds that "Flaviviruses and Orthomyxoviruses may be strong candidates for inclusion" subject to resources — so two of the three gaps are declared *and* queued, while Picornaviridae is declared and not queued.

Regional burden, measured directly. Africa CDC has run the same MCDA-plus-Delphi instrument separately in each region and published the divergent results. Coronaviruses rank **first** in Northern Africa (listed as "coronavirus diseases", plural, at family level), **fourth of 22** in Eastern Africa (as "COVID-19", risk score 9.9, behind Ebola, Marburg and cholera), and **outside the published top five** in Southern Africa. Influenza, by contrast, is top-five in all three. Pulik et al. observe the same pattern globally: influenza outranks rabies in high-income countries and the reverse holds in low- and middle-income ones. HERA deprioritises Rhabdoviridae outright — the family containing the single most frequently prioritised pathogen worldwide — while Japan places rabies in Group D as an imported threat. One organism, three different answers, each defensible from its own geography.

Date of revision. The WHO legacy "priority diseases" list, which is still live, explicitly placed "influenza causing severe human disease" *outside* the Blueprint's scope. The 2018 annual review is unambiguous: "Several diseases were determined to be outside of the current scope of the Blueprint: dengue, yellow fever, HIV/AIDS, tuberculosis, malaria, influenza causing severe human disease, smallpox, cholera, leishmaniosis, West Nile Virus and plague." WHO's 2024 framework includes Orthomyxoviridae with seven priority subtypes. The WHO R&D Blueprint's own 2022 concept note tabulated five frameworks and found influenza present on exactly **one**. Four years later it is on WHO, UKHSA, HERA and PHEMCE but still not CEPI. The trajectory is the finding, and the WHO 2018 column in the matrix above now makes it visible: four of the matrix's eighteen families (Orthomyxoviridae, Flaviviridae, Poxviridae and Retroviridae) carried a 2018 entry only as a declared *exclusion*.

Procurement versus research framing. The EU authorises four pandemic-preparedness "mock-up" influenza vaccines with a strain-change procedure and three zoonotic/pre-pandemic vaccines — a regulatory and stockpiling track with no analogue for any other family. PHEMCE's five-year need for pandemic influenza is $15,654m against a total estimated five-year need of $66,888m, or **23%** (MYB Appendix C, p.41; the total is also stated in the executive summary, p.1: "the PHEMCE projects an estimated overall funding need of $66.9 billion"). Yet the same influenza is largely unregulated as laboratory material, and NIAID leaves it out of its prototype programme. A naive concordance count across these lists would be meaningless.

Unit of analysis. This is the driver the matrix makes hardest to see and easiest to mistake for substantive disagreement. The frameworks prioritise at six different ranks: subgenus (WHO's coronavirus prototypes), HA subtype (WHO's influenza priorities), species (NIAID's priority track, Select Agents), genus (WHO's *Vesiculovirus*, *Enterovirus*), family (UKHSA, HERA, PHEMCE's EID bucket, CEPI from 2027, Adalja & Inglesby) and disease (Africa CDC, Japan MHLW, WHO's legacy list). A tick and a blank in adjacent columns of the matrix above can therefore mean "these bodies disagree about this family" or merely "one of them does not have a cell at this rank". Three concrete cases: Africa CDC's Northern region lists "coronavirus diseases" at family level while its Eastern region

lists "COVID-19" at disease level, using the same instrument — so the apparent regional divergence is partly a unit shift; PHEMCE's Filoviridae and Poxviridae cells are double-booked, appearing both in the family bucket and as MTD-asterisked named agents (Ebolavirus, Marburgvirus, Variola), a dual-track structure no other framework has; and WHO declined to extend the prototype concept to bacteria for an explicitly unit-based reason — "Prototype pathogens were not recommended for bacteria, because of the uniqueness of each priority pathogen." Pulik et al. reached the same conclusion from a larger sample: their "ability to synthesise a meaningful ordered priority list across publications was limited by discrepancies in taxonomy and scope". Any use of the matrix above should check the rank before reading a blank as a disagreement.

A first count, and what it does not show. The claim that purpose is the single biggest source of divergence is asserted throughout this note and has not previously been measured. A first pass is possible, with a stated rule: count every family–framework cell in which a framework omits or deprioritises a family that at least one other framework places in its top tier, *and* for which the omitting framework publishes a rationale, then attribute each to the rationale given.

DRIVER	PUBLISHED-RATIONALE OMISSIONS	WHICH
Criteria threshold (one framework's shared rationale)	7	HERA's seven explicitly non-prioritised families, all covered by one sentence
Purpose / existing machinery	6	WHO 2018's out-of-scope declarations touching Orthomyxo-, Flavi-, Pox- and Retroviridae; UKHSA's abstention from the prototype layer; PHEMCE's MTD gating
Mandate / market failure	3	CEPI × Orthomyxoviridae, Flaviviridae, Picornaviridae
Resourcing as an inverse criterion	2	NIAID × Coronaviridae, Orthomyxoviridae
Region / burden	1	HERA's exclusion of measles
Revision date	1	HERA's Peribunyaviridae deferral
Unit of analysis	1	WHO's refusal to extend prototypes to bacteria

The count does **not** support "purpose is the biggest source" on raw cells: the largest bucket is a single framework's threshold rationale, which is an artefact of HERA happening to publish one sentence covering seven families at once. It supports the claim only if cells are weighted by consequence, in which case the two NIAID resourcing cells and the three CEPI mandate cells carry far more weight than seven HERA rejections of families nobody else prioritises either. The honest reading is that purpose is the biggest source of *consequential* divergence and the count makes that dependence on weighting explicit rather than hiding it. The coding above is this note's own and is reproducible from the matrix; it is not a published result and should not be cited as one.

One organism, two current frameworks, opposite calls. The cleanest single instance in the whole landscape is measles. HERA excludes it by name, with a published reason: "*Measles have been excluded because outbreaks within the EU/EEA result from reduced vaccination uptake, not intrinsic cross-border threat from viral evolution or emergence*" (SWD(2026) 1 final, figure note, p.18). The WHO Paramyxovirus R&D Roadmap of 31 March 2026 names measles as one of seven prototype viruses for the family. Both documents are current, both are dated within three months of each other, both are from bodies that cite each other, and they reach opposite conclusions about the same organism — because one is asking whether measles is a cross-border MCM-procurement threat in the EU and the other is asking whether measles is a good model organism for paramyxovirus countermeasure development. Neither is wrong. This is what the rest of this section is about, compressed into one example.

(c) Notable absences, and why

Caliciviridae is not among the families WHO assessed at all, yet UKHSA rates it Medium/High on norovirus and HERA names it explicitly as not prioritised. **Retroviridae** is a WHO Medium-risk family whose priority pathogen and prototype are both HIV-1, is absent from UKHSA and CEPI, and is not prioritised by HERA or named by Adalja & Inglesby. HERA's reason for Retroviridae is *not* specific to it: the single sentence covering all seven non-prioritised families says they "do not currently meet the criteria for higher epidemic and pandemic potential (including sustained human-to-human transmission, rapid cross-border spread and limited availability of effective countermeasures) and were therefore not captured by the applied filtering approach." There is no Retroviridae-specific transmission-mode rationale in the document, and this note previously implied there was. **Pneumoviridae** is on Adalja & Inglesby's six and UKHSA's list but HERA does not prioritise it. **Rhabdoviridae** is the sharpest case of all and now has its own matrix row: WHO rates the family Low and proposes no priority pathogen while naming *Vesiculovirus* a prototype on pure tool-value grounds, HERA rejects it outright, UKHSA does not list it, and it contains rabies — joint first in the 13-resource empirical record. **Mycobacteriaceae** is absent from UKHSA's bacterial families, which is the omission that drew the sharpest published criticism; note that WHO does list drug-resistant *M. tuberculosis* in its Bacterial Priority Pathogens List, so the gap is UKHSA's rather than the field's. **No coronavirus of any kind** appears on WHO's Immunization Agenda 2030 endemic-pathogen list — a scoping artefact, since that exercise addressed endemic pathogens and drew its quantitative criteria from GBD 2019 data that predate COVID-19.

And one absence that matters for the implementation layer: **Orthomyxoviridae is one of two High-risk WHO families with no CORC and no family R&D roadmap.** The other is **Togaviridae**, which this note's own matrix rates WHO High with chikungunya and VEEV as both priority and prototype. The 7 April 2026 launch page lists CORCs for Arenaviridae, Bunyavirales, Coronaviridae, Filoviridae, Flaviviridae, Paramyxoviridae and Poxviridae, plus one bacterial consortium; neither Orthomyxoviridae nor Togaviridae appears. Orthomyxoviridae had appeared as a row in WHO's February 2025 "Future CORCs" table with the lead-institution cell blank. No WHO statement explains either omission.

This matters for how the omission should be read. The intuitive explanation for influenza — that it is governed by separate long-standing machinery (GISRS, the PIP Framework, the Global Influenza Programme) — is **unverified**; the 2024 framework does not mention any of those mechanisms. And it cannot be the whole story, because it does not explain Togaviridae, which has no parallel machinery. The inference that influenza's absence from the CORC layer signals deliberate special handling should therefore be dropped, or at most held as one of two hypotheses alongside the duller one: that the CORC programme is incomplete and these two families have not yet been allocated a host.

4. The prototype concept in depth

Origin. The term enters policy through Barney Graham and Nancy Sullivan's *Nature Immunology* review (online 14 December 2017), which proposed that after sequencing, viruses "could be further categorized according to entry mechanism, tropism or route of transmission to select a finite number of representative viruses to serve as prototypes". Graham and Corbett (2020) sharpened the rationale: because "virus families and genera are organized based on shared functional and structural properties, vaccine solutions within viral families and genera may also be similar". NIAID operationalised it in December 2021; WHO borrowed it in 2024. Note a correction worth having: Graham and Sullivan's Table 2 *does* assign prototypic viruses to Orthomyxoviridae (influenza A and B), so the foundational paper did not anticipate NIAID's later exclusion of that family.

The structure-based-design argument. The claim is that taxonomy predicts structural and antigenic similarity, so a stabilisation strategy, an immunogen-design rule, an assay suite, a reagent panel or an animal model built against one family member transfers to the rest. Graham and Sullivan put the strongest version at the level of protein class rather than taxon: the design process "might be the same for any envelope glycoprotein that functions as a trimeric class I fusion" protein, regardless of virus. WHO states the molecular version explicitly and in terms that map directly onto small-molecule and peptide discovery: "For antivirals, inhibitors of viral enzymes that are similar within the family may be effective against multiple family members. For vaccines, this knowledge may facilitate understanding of viral antigens likely to induce protective immunity or likely correlates of protection."

The evidence that tools transfer, and its limits. The evidence base is a small set of worked case histories, not a systematic test. The load-bearing one is coronaviruses: the two-proline prefusion-stabilising substitutions were established in MERS-CoV spike by Pallesen et al. (2017), building on McLellan et al.'s RSV prefusion F work (2013), and carried directly into SARS-CoV-2 S-2P — published by Corbett et al. (2020) under the title *SARS-CoV-2 mRNA vaccine design enabled by prototype pathogen preparedness*. Secondary cases cited are the acceleration of Zika candidates from prior flavivirus work, and Ebola GP safety data facilitating other glycoprotein-vectored vaccines.

Three limits should be stated plainly.

First, **no peer-reviewed study empirically testing whether countermeasure tools generalise across a viral family was found in this survey**. This is a bounded negative, not a proof of absence: the search behind it was iterative keyword and citation-chasing across PubMed/PMC, Google Scholar and the reference lists of the sources cited in this note, conducted during October 2026; no formal protocol, database set or search string was pre-registered, and no systematic-review method was applied. It is stated in that weaker form everywhere it appears below. Anyone relying on it should re-run the search properly, because three separate arguments in this note depend on it.

Second, the proof of concept is partly circular — it validates the approach by pointing to the family that was already the best-resourced candidate and that NIAID then excluded for exactly that reason. Morabito et al., *Viral Prototypes for Pandemic Preparedness: The Road Ahead*, *J Infect Dis* 2023;228(Suppl 6):S460–S466 concede both the success and the asymmetry: COVID-19 is "proof of concept that the prototype pathogen approach is feasible", but "for other virus families of pandemic concern, the field is less prepared", and "because not all [concepts transfer] to related viruses, intensive study of secondary viruses within the family will be required to validate generalizable MCM development approaches".

Third, counter-evidence exists from inside the programme: Nancy Sullivan, co-chair of the Filoviridae working group, presented a slide to a WHO consultation titled "Filoviruses: 'Prototypes' may not protect across virus family", concluding that prototype choice "should account for differences in immune clearance mechanisms". **This single slide is the only direct counter-evidence to the transfer premise located in this survey, and it does a great deal of work in this note and in section 6.** Its evidentiary weight should be stated honestly: it is an unrefereed presentation slide from a consultation deck, by a credible insider, with no accompanying published paper found. It is enough to establish that the transfer premise is contested inside the programme. It is not enough to establish that transfer fails, and it should not be cited as if it were a result.

How prototypes are actually selected. This is the part most relevant to section 6, and it is more favourable to tractability arguments than the rhetoric suggests. WHO's published criteria are "importance as human pathogens, current knowledge of replication and pathogenesis, the existence of animal reservoirs causing cross-species infections, the shared structural and functional properties, the existing research knowledge, for example, the availability of animal models that recapitulate human disease, and the status of countermeasure development", with ties broken on "burden and type of disease, existing collaborations, and reagents are likely to speed up work on one pathogen or another". Hartman and Myler, writing for the Bunyavirales working group, state three criteria: prototypes should "encompass the breadth of clinical outcomes of a given family", "have existing reverse genetics tools or animal disease models", and "can be amenable to a platform approach to vaccine testing". NIAID's Table 1 rationales include "reagents available" (EV-A71), "well studied" (rhinovirus C), "surrogate that can be studied at BSL-2" (Hazara) and "less lethal; BSL-3" (Hantaan). WHO allowed multiple prototypes per family where diversity made one insufficient, and declined to extend the concept to bacteria at all: "Prototype pathogens were not recommended for bacteria, because of the uniqueness of each priority pathogen."

Neither WHO nor NIAID published a scoring rubric, criterion weights, voting rule or per-family score. Both exercises are structured expert elicitation. There is a formal alternative in the literature that nobody in the policy space has adopted: PARNAS frames representative selection as a generalised k-medoids problem solved exactly on a phylogeny, permits weighting taxa by metadata and constraining the candidate pool, and was built for influenza A surveillance and vaccine design. It is the strongest published answer to "how would you choose a prototype non-subjectively".

Published critiques, and what is advocacy versus evidence. Most of what is written about the prototype concept is advocacy by its originators and implementers, plus self-stated caveats. The genuinely external critiques are thin and mostly aimed at prioritisation generally rather than at prototypes specifically. Pulik et al. (2026) found that across 13 exercises, several transparency practices "were never followed" (human and financial resources identification, evaluation plan and process, monitoring of impact and change, dissemination), that "the methods were all unclear and not reproducible", that a qualitative discussion step overrode the quantitative stage in 10 of 11 exercises, and that **no resource involved policy-makers or research funders as participants**. Antonio et al. (2025)

found that only 6% of 125 research-prioritisation publications indicated plans to monitor progress against the priorities they set. Monrad et al. (2021) argue the economic case that pathogen-specific funding structurally underfunds exactly this kind of generalisable work. Christofferson and Cormier (2022) argue the Disease X framing is too narrow. And CSET's 2023 analysis supplies the quantitative point that matters most here: pandemic-potential viral families "make up only 14 percent of all virus-related research publications", publication counts "peak after major outbreaks", and the post-2020 coronavirus corpus is "skewed".

One stability warning. The 2024 prototype selections are already being revised downstream. The Paramyxovirus R&D Roadmap of 31 March 2026 names **seven** prototype viruses (measles, Nipah, HPIV-3, Cedar, Menangle, PIV-5, Newcastle disease virus) where the 2024 framework named one. Citing the 2024 table as "the" WHO prototype list, without checking the relevant family roadmap, is already out of date for at least one family.

5. Coronaviridae and Orthomyxoviridae specifically

Both tables below use the same column headers, so they can be read side by side.

Coronaviridae

FRAMEWORK	STATUS	PRIORITY UNIT	PROTOTYPE
WHO 2024	High PHEIC risk; "multiple pandemic threats in family"	**Subgenus** Sarbecovirus (all six regions), subgenus Merbecovirus (EMR only)	The same two subgenera. **No species-level prototype.** Note: "surveillance should consider animal reservoirs" is the only published prototype rationale, identical for both
NIAID	Excluded from the prototype exercise by name; handled as a priority pathogen plus the Antiviral Program for Pandemics	SARS-CoV-2 as an example priority pathogen	**None**
UKHSA	High / High	**MERS-CoV only**. SARS-CoV-2 is not named in any table	Declines the concept
CEPI	Standing priority; "up to US$ 214.5 million in a portfolio of 14 BPCV vaccines projects, six of which are active" (2025 Annual Progress Report, Coronaviruses section). *Caution:* CEPI's live broadly protective coronavirus vaccines page gives "up to US$230 million" across the same 14 projects, and a November 2025 CEPI graphic says 13 candidates. Use the report figure and note the discrepancy	MERS-CoV	**MERS-CoV**, with a separate pan-sarbecovirus track
HERA	Group 1, highest priority; "very high" EU cross-border threat potential	SARS-CoV, SARS-CoV-2, MERS-CoV; also notes WHO has identified sarbecoviruses as priority pathogens	Imports WHO's two subgenera
PHEMCE	EID priority family, second on the list; $645m over FY2025–29 (MYB Appendix C, p.41)	SARS-CoV-2 (SARS-CoV appears with an MTD asterisk in the 2024 SIP appendix but is absent from the current standalone threats page — unresolved, see section 7)	None

Two things are worth drawing out. First, **WHO prototypes this family at subgenus level**, not species level. That is the finest-grained warrant anywhere in policy for working against a conserved target shared across a clade rather than against one named virus. Second, **UKHSA's choice of MERS-CoV over SARS-CoV-2 is a countermeasure-gap choice**, not a threat choice: its row records "Nil specific / Supportive only / Candidates in phase 2 trials" for therapeutics and no licensed vaccine. HERA records the same gap at EU level in section 4.1.4 (p.31): "There are currently no approved vaccines or therapeutics for MERS-CoV, SARS-CoV-1 or endemic human coronaviruses (HCoV-229E, HCoV-OC43, HCoV-NL63 and HCoV-HKU1) as of October 2025", and its family-overview figure (p.18) records "no universal" vaccine and "no broad-spectrum" antiviral for the family. CEPI's rolling Focus Area 2 call, in its April 2026 revision, scopes its Disease X exemplar-vaccine track to "Coronaviruses **other than** ACE-2 binding Sarbecoviruses with human spillover or potential for human spillover that pose a credible epidemic or pandemic threat", alongside paramyxoviruses, arenaviruses, phenuiviruses and nairoviruses — sarbecoviruses being covered by the separate pan-sarbecovirus programme. The coronavirus white space CEPI was buying was non-sarbecovirus spillover-capable coronaviruses.

Orthomyxoviridae

FRAMEWORK	STATUS	PRIORITY UNIT	PROTOTYPE
WHO 2024	High PHEIC risk	**Seven HA subtypes**: H1N1, H2Nx, H3N2, H5Nx, H6Nx, H7Nx, H10Nx. H9N2 and influenza B are Pathogen X candidates	**Two**: (H1N1), noted "human"; (H5Nx), noted "avian". Five of the seven priority subtypes are not prototypes
NIAID	Excluded by name; handled through CEIRS/CEIRR and the universal influenza vaccine programme	"Influenza virus" as an example priority pathogen, no subtype	**None anywhere in the framework**
UKHSA	High / High	"Non-seasonal influenza". No subtype named. HCID column reads "Avian"	Declines the concept
CEPI	**Not a priority pathogen.** Named as a strategic gap; flagged in Investment Case Annex 2 as a "strong candidate for inclusion" subject to resources	"Pandemic Flu" appears in CEPI's access-archetype deck footnoted "Not a CEPI priority pathogen"	Influenza is a *platform-validation* prototype (a different sense entirely); "Commitment of up to US$ 54.3 million to support the pivotal Phase 3 clinical trial of Moderna's investigational mRNA-based H5 pandemic influenza vaccine candidate" as a stated exception (2025 Annual Progress Report, p.30)
HERA	Group 1; "very high" EU cross-border threat potential	Restates WHO's seven subtypes verbatim	Restates WHO's H1 and H5
WHO EMRO	**Names A(H5N1) directly in its priority list** — the only framework surveyed to do so in the list proper	A(H5N1) plus "influenza caused by any novel and zoonotic viruses"	n/a
PHEMCE	Standalone priority threat **with no MTD**; also the EID family listed first; $15.65bn over FY2025–29 (MYB p.20, Appendix C p.41)	"Pandemic influenza"; H5N1 and H7N9 named in funded activities	None

The influenza carve-out is real, documented, and changed direction in 2024. The 2018 WHO annual review listed influenza explicitly as out of scope: "Several diseases were determined to be outside of the current scope of the Blueprint: dengue, yellow fever, HIV/AIDS, tuberculosis, malaria, influenza causing severe human disease, smallpox, cholera, leishmaniosis, West Nile Virus and plague. These diseases continue to pose major public health problems and further research and development is needed through existing major disease control initiatives, extensive research and development pipelines, existing funding streams, or established regulatory pathways for improved interventions." The Orthomyxoviridae row in the 2024 framework's own comparison table is blank for both 2017 and 2018, and HERA's Figure 1 confirms it independently: its dated "R&D blueprint 'priority pathogens' 2018" column carries no Orthomyxoviridae tick. Influenza's entry in 2024 is the single largest addition of that revision.

Why this matters. Every major body operates parallel influenza machinery that sits outside its general prioritisation: WHO's GISRS and PIP Framework and a separate Public Health Research Agenda for Influenza; NIAID's CEIRS/CEIRR and the universal influenza vaccine plan; the EU's four authorised mock-up vaccines with an EMA strain-change procedure plus three zoonotic vaccines; PHEMCE's dedicated pandemic influenza appropriation outside Project BioShield; BARDA's whole Influenza and Emerging Infectious Diseases division. The result is that influenza simultaneously carries the most funding, the widest subtype-level specificity in the lists that do include it, and the weakest presence in the *prototype* layer: NIAID names no influenza prototype, CEPI names none, UKHSA names none, and WHO's two (H1N1 and H5Nx) are split on the human/avian axis and nothing else.

Orthomyxoviridae is also one of two High-risk WHO families with no CORC and no roadmap, the other being Togaviridae. As set out in 3(c), that pairing is the reason the "separate machinery" explanation cannot be carried as an inference: Togaviridae has no separate machinery and is missing too.

6. Implications for a target-selection argument

The project selected Coronaviridae via SARS-CoV-2 Mpro, and Orthomyxoviridae via influenza haemagglutinin stem and neuraminidase, substantially because public deep-mutational-scanning measurement data exist for those targets and unblock method development. Is "tractability and data availability" a defensible prototype-selection criterion in the published literature, or a convenience argument dressed up?

This section is ordered so the answer comes last. The case for, the case against, the counterfactual and the falsification conditions are set out first, and the verdict at the end is contingent on them. Readers who want the conclusion should note that it is a conditional one and that the conditions are not currently met.

6.1 The strongest version of the case for

Tractability is not merely tolerated in this literature; it is written into the published criteria. WHO's own prototype criteria include "existing research knowledge", animal-model availability, and — as a tie-breaker — whether "existing collaborations, and reagents are likely to speed up work on one pathogen or another". Hartman and Myler require that a prototype "have existing reverse genetics tools or animal disease models" and "be amenable to a platform approach". NIAID's published rationales include "reagents available" and "well studied". WHO selected vaccinia *because* it offers a BSL-2 route and *Vesiculovirus because* it is a useful vector. These are feasibility criteria embedded inside public-health priority lists, and they are not hidden — they are printed in the tables. A project that says "we chose this target because measurement data exist that let us validate the method" is applying the same class of criterion that the WHO Prioritization Advisory Committee applied.

The two families are also unimpeachable on the risk axis. Both are High PHEIC risk at WHO, both High/High at UKHSA, both Group 1 at HERA, both in PHEMCE's emerging-infectious-disease bucket (listed first and second), both on Adalja & Inglesby's narrowed six, and influenza is jointly the most frequently prioritised pathogen in the entire 13-resource empirical record. Both are simultaneously WHO priority *and* prototype families — so the project is not working outside the prototype layer, it is working inside it.

The published countermeasure gaps are specific, quotable and now sourced to page. For Coronaviridae: HERA section 4.1.4 (p.31) records that "There are currently no approved vaccines or therapeutics for MERS-CoV, SARS-CoV-1 or endemic human coronaviruses (HCoV-229E, HCoV-OC43, HCoV-NL63 and HCoV-HKU1) as of October 2025", its family-overview figure (p.18) records "no universal" coronavirus vaccine and "no broad-spectrum" coronavirus antiviral, and UKHSA records "nil specific / supportive only" for the family's exemplar. For Orthomyxoviridae, HERA states at p.19: "Influenza viruses evolve rapidly through antigenic drift and shift, which allows them to escape host immunity and limits the long-term effectiveness of vaccines. Seasonal influenza vaccine effectiveness varies between strains and seasons, but the effectiveness of neuraminidase inhibitors and baloxavir has remained constant stable, with only sporadic reports of reduced susceptibility. By contrast, widespread resistance has rendered adamantanes clinically obsolete." Section 4.2.4 (p.39) gives the mechanism and the scope: adamantanes "have now been effectively abandoned because of the near-universal resistance among the currently circulating human influenza A(H1N1)pdm09 and A(H3N2) viruses... primarily mediated by the S31N substitution in the M2 protein", and resistance "is also common in zoonotic influenza viruses". CEPI names the family as a self-declared, unfilled gap. A therapeutic-peptide programme is aimed squarely at those gaps.

Note what that HERA passage does *not* say, because the distinction matters for the project's framing: HERA records the small-molecule antiviral repertoire for influenza as narrow but *stable* — neuraminidase inhibitors and baloxavir are working. The gap HERA identifies is in vaccines, through antigenic variability, not in antivirals. A programme justified as filling an influenza *therapeutic* gap is making a claim HERA does not support; a programme justified as filling the breadth gap — tools that survive antigenic change — is making one HERA does.

Finally, the unit of prioritisation supports the choice. WHO's Coronaviridae prototype is a **subgenus**, not a species — which is precisely a clade of related proteins. Mpro is among the most conserved enzymes across that clade and beyond it. WHO's own molecular rationale for the family approach is that "inhibitors of viral enzymes that are similar within the family may be effective against multiple family members". The project's targets sit exactly where WHO's argument says the transferable work is. Section 3(b) establishes that unit mismatch is a general problem across these frameworks, which is what makes this a real argument rather than a convenient one: the project is matching WHO's stated rank rather than silently substituting a different one.

6.2 The strongest version of the case against

The decisive objection is that **the one framework that formally operationalises prototype selection excluded both of these families, by name, for a reason that applies directly here**: "significant resources are already allocated to characterize these families, and there are licensed or advanced clinical stage development MCMs available for viruses within them." NIAID's criteria (2) and (3) are inverse — the better-resourced a family, the lower it ranks. A reviewer aligned with NIAID will not ask whether the choice is risky enough; they will ask why not Arenaviridae, Nairoviridae or Picornaviridae, where ReVAMPP is actually spending and where the field is demonstrably unprepared. UKHSA independently reaches a compatible position by anchoring Coronaviridae on MERS-CoV rather than SARS-CoV-2.

Second, and more corrosive: **a data-availability criterion partly encodes outbreak history, not tractability or representativeness.** CSET found that research publications and clinical trials peak after outbreaks, that pandemic-potential families account for only 14% of virus-related publications, and that the post-2020 coronavirus corpus is "skewed". Public deep-mutational-scanning data exist for SARS-CoV-2 spike and Mpro, and for influenza HA and NA, precisely because these are the two most intensively studied viral families in history following a pandemic and a century of influenza programmes. Selecting on data availability therefore selects on past attention. That is a real circularity and the reviewer will see it immediately.

Third, **representativeness and tractability are different criteria and the frameworks treat representativeness as primary**. RFA-AI-23-020 lists, in order, how well the prototype reflects shared family traits; whether one or several prototypes are needed; experimental tractability; and public-health need. Tractability is third. Nothing in any published criterion says "pick the family with the most data"; the criteria say "pick a representative organism, provided you can work on it". Data abundance is evidence of the enabling condition, not of the selection criterion.

Fourth, **within-family transfer remains asserted rather than demonstrated** — no systematic test was found in this survey (see section 4 for what that search was and was not) — and Adalja & Inglesby make the specific point that cross-protective countermeasures do exist within orthopoxviruses but do *not* exist for "the pandemic-causing orthomyxoviruses and coronaviruses". These are the two families where immunological transfer within the family is weakest. Tool transfer and immunity transfer are different claims; only the first is well supported here, and conflating them would be a visible error.

Fifth, a precision point. WHO's Orthomyxoviridae prototypes are H1N1 and H5Nx. "Influenza haemagglutinin stem" is not a WHO unit, and HA group 1 versus group 2 conservation is not the same question as H1-versus-H5. The project needs to say which subtype's HA, and whether the claim is intra-group or cross-group.

6.3 The counterfactual: what choosing an under-resourced family would actually cost

The case against has a specific alternative behind it, and until now this note has not priced it. NIAID's ReVAMPP network commits approximately US$100m per year across nine families — Flaviviridae, Paramyxoviridae, Picornaviridae, Togaviridae, Arenaviridae, Hantaviridae, Nairoviridae, Phenuiviridae and Peribunyaviridae. Three named awards are directly relevant: PABVAX (Geisbert, UTMB) on Paramyxoviridae and Bunyavirales, BP4 (Crowe, Vanderbilt) on Bunyavirus and Picornavirus, and PROVIDENT (Chandran, Einstein). Three concrete alternative targets, and what is actually available for each:

Lassa virus glycoprotein complex (Arenaviridae). WHO priority pathogen and prototype; CEPI priority with a Phase 3 trial committed in CEPI 3.0; ReVAMPP-funded. *Quantitative public data that exist:* multiple solved prefusion GPC structures, a substantial published monoclonal antibody panel with neutralisation measurements, and lineage-diversity sequence sets from West African surveillance. *What does not exist:* no published deep mutational scan of GPC at the scale of the SARS-CoV-2 or HA datasets, and no comparable public binding-affinity matrix. A method needing dense per-variant quantitative labels cannot be trained or honestly evaluated here today.

CCHFV glycoprotein or L-protein OTU domain (Nairoviridae). WHO priority pathogen and prototype; ReVAMPP-funded through BP4; NIAID additionally names Hazara virus as a BSL-2 surrogate with an existing reverse-genetics system, which is exactly the tractability move the frameworks endorse. *Quantitative public data that exist:* structures, a growing antibody literature, and the Hazara surrogate system. *What does not exist:* no public DMS, and the BSL-4 requirement for CCHFV itself means that generating one is not a matter of effort but of facility access.

Enterovirus A71 or D68 capsid (Picornaviridae). WHO prototypes for the family; ReVAMPP-funded; and the most tractable of the three — EV-A71 and EV-D68 are BSL-2, NIAID's own stated rationale for EV-A71 is "reagents available", and the enterovirus capsid has a long structural and antiviral-chemistry literature including pocket-binder series. *Quantitative public data that exist:* structures, antiviral SAR series, some mutational-tolerance work. *What does not exist:* no DMS dataset of comparable density and no standard held-out benchmark.

The honest cost of switching. For Picornaviridae the cost is moderate and mostly a matter of assembling a benchmark rather than generating one from scratch — structural and chemical data exist, and BSL-2 access is not a barrier. For Arenaviridae and Nairoviridae the cost is high and partly non-monetary: generating the quantitative labels a learned method needs would require either BSL-4 access or surrogate systems whose relationship to the real target is itself an open question, which is a research programme in its own right and not a target substitution.

What this does and does not settle. It settles that the claim "these two families are the only place the method can be validated" is too strong, and this note previously assumed it rather than arguing it. Quantitative public data of *some* kinds — structures, antibody panels, antiviral SAR — exist for all three alternatives, and for Picornaviridae the gap to a usable benchmark is a matter of assembly work rather than a wall. What it does not settle is the narrower and more defensible claim, which is the one the project should actually make: *dense per-variant fitness and binding measurements of the kind required to train and honestly evaluate a learned predictor, with enough held-out structure to test generalisation, exist today only for SARS-CoV-2 and influenza targets.* That claim is true as far as this survey could establish, it is falsifiable, and it is narrower than "the only place the method can be validated". The project should make that claim and not the broader one.

6.4 What would have to be true for the choice to be defensible to a reviewer

1. **The selection criterion must be stated honestly and separated from the representativeness argument.** Say: these families were chosen because validated public measurement data make method development and honest evaluation possible; the representativeness case is argued separately, on target conservation, and does not depend on the data argument. Do not let the two merge.

1. **Generalisation must be demonstrated out of distribution, not asserted.** Cross-validation within SARS-CoV-2 is not evidence of transfer. Held-out evaluation on a related but unseen target — another sarbecovirus Mpro, a merbecovirus Mpro, a different HA subtype, ideally across HA groups — is the minimum. If the method only works where DMS data already exist, it is a fitting exercise, not a preparedness method.

1. **Target conservation must be established at the taxonomic level the frameworks actually prioritise.** WHO prioritises Coronaviridae at subgenus level, so the Mpro conservation argument should be made across Sarbecovirus and Merbecovirus, not against SARS-CoV-2 alone. WHO prioritises influenza at HA-subtype level, so the stem and NA arguments should be made across the named priority subtypes.

1. **NIAID's exclusion must be answered on its own terms.** The answer available is that NIAID's exclusion concerns *product* resourcing for countermeasure development, not benchmark availability for *method* validation; a family with dense, public, quantitative binding and fitness measurements is the correct place

to validate a method whose intended destination is the resource-poor families. That answer is not automatically available. It holds only if point 2 is actually delivered, if the eventual application to a data-poor family is stated as the goal rather than left implicit, and if 6.3's narrower claim survives contact with someone who knows the Picornaviridae data landscape better than this survey does. If it does not survive, the exclusion stands and the choice should change.

1. **The selection should ideally be formalised.** PARNAS exists, solves representative selection exactly on a phylogeny, and allows the candidate pool to be constrained and taxa to be weighted by metadata. Encoding "has public DMS data" as a *constraint* on the candidate pool and then optimising for phylogenetic coverage converts a convenience argument into a constrained optimisation with a stated objective. That is a materially stronger position than any of the policy bodies occupy, and it is the one methodological move available that a reviewer could not characterise as post-hoc.

6.5 Two falsification conditions, named in advance

Falsifier for transfer. State what result would show that the method does not transfer — for instance, performance on a held-out merbecovirus Mpro or a group-2 HA indistinguishable from a sequence-identity baseline. If that is the result, the preparedness framing fails and the work is an interpolation exercise on two well-measured proteins.

Falsifier for the target choice itself. The data-availability justification is a claim about the world, not a preference, and it expires if the world changes. State the condition: *if a benchmark of comparable density and held-out structure can be assembled for a target in an under-resourced family — Picornaviridae is the most likely candidate on the analysis in 6.3 — within a stated time budget of (say) four to six weeks of one person's effort, then the data-availability justification lapses and the target choice should be revisited.* This is a real falsifier because it can be run cheaply and because, on 6.3's own reading, Picornaviridae is close to the line. Running it before Thursday is not possible; committing to run it, and saying what the answer would mean, is. If the benchmark can be assembled, the defensible choice is to assemble it and work there, because that is where both the NIAID criterion and the public-health need point.

6.6 Verdict

Tractability and data availability are defensible prototype criteria with direct textual support in WHO's and NIAID's published selections. But they are *secondary* criteria in every framework that states an ordering, and in these two specific families they coincide almost perfectly with the resourcing that caused NIAID to exclude them.

So the verdict is conditional, and the conditions are the ones above. **The choice is defensible if and only if the narrow data claim in 6.3 holds, point 2's out-of-distribution evaluation is delivered rather than promised, and the 6.5 falsifier is run and comes back negative.** None of those three is currently satisfied. As things stand on 20 October 2026, the choice is *reasonable and unproven* — which is a different and weaker position than this note previously claimed, and it is the position the evidence actually supports. Presented as though the frameworks simply endorse it, it will not survive a reviewer who knows the NIAID exclusion. Presented as a conditional with the conditions named and a date attached to testing them, it is a stronger paper than one that claims more.

7. What is unresolved

Whether the final Coronaviridae R&D roadmap is indexed on WHO's publication pages. The substantive question this note previously raised here is now closed: Coronaviridae **does** have a CORC, hosted by PREPARE Singapore, per the 7 April 2026 WHO announcement, and the roadmap document itself (CORC-CoV, first draft 12 January 2026, public consultation completed 28 February 2026, revised 1 April 2026) is reachable at `cdn.who.int`. What remains open is only bibliographic: it is not indexed on WHO's publication pages the way the Filovirus, Arenaviridae and Paramyxovirus roadmaps are. *Settled by:* a who.int publication-page listing, or a WHO statement of its publication status. This still matters, because the Paramyxovirus roadmap revised its family's prototype list from one to seven, so the 2024 subgenus-level coronavirus prototypes may no longer be operative.

Why Orthomyxoviridae and Togaviridae have no CORC and no roadmap. No WHO statement addresses either. The GISRS/PIP explanation for influenza is an inference, the 2024 framework mentions none of those mechanisms, and it does not extend to Togaviridae, which has no parallel machinery. *Settled by:* a WHO Blueprint statement, or correspondence with the Blueprint secretariat.

Whether SARS-CoV has been dropped from the PHEMCE high-priority threat list, or the two ASPR pages are simply unsynchronised. The September 2024 SIP appendix carries "Severe acute respiratory syndrome-related coronavirus (SARS-CoV) (including SARS-CoV-1 and -2)" with a Material Threat Determination asterisk; the current standalone Priority Threats page omits it (re-checked 20 October 2026); and the July 2026 Multiyear Budget's enumeration of MTD threats in the biological portfolio does not include it. Note that this is a question about PHEMCE only — the separate Select Agents position is now verified and is *not* uncertain (see below). *Settled by:* a current public DHS Material Threat Determination roster, which does not appear to exist in published form.

How many viral families WHO actually assessed. The prose says 28; Tables 1 and 2 enumerate 29; Table 14 adds a thirtieth (Arteriviridae) that appears in no risk table; the PAC section implies 27 (8 DNA + 19 RNA); Annex 1 groups them into roughly 25 expert groups because Bunyavirales and Reovirales were each run as one. No single figure reconciles. *Settled by:* a WHO erratum, or by counting from the tables and saying so explicitly rather than quoting the headline.

Whether UKHSA has issued a 2026 edition. The GOV.UK page still shows March 2025 / 8 April 2025 — and states that the April update was a page-11 edit reflecting the change in HCID status of clade I mpox, which bears directly on this note's Poxviridae row ("Medium / High — clade I MPXV") and on section 2's claim that UKHSA's HCID treatment is part of its distinctiveness. But a UKHSA blog of 9 March 2026 states the tool "now covers 24 **viral** families", which does not match the published 16 viral plus 8 bacterial. Either the blog is loose, or an expanded tool exists outside the GOV.UK publications system. *Settled by:* UKHSA confirmation, or a retrieved 2026 PDF.

Which of WHO's nine post-2017 priority pathogens CEPI does not support. The CEPI 3.0 Investment Case (p.6) commits to "eight of the nine viruses designated priority pathogens by the WHO since 2017, and against 'prototype pathogens' from at least nine of the 11 viral families designated by the WHO as presenting a high risk", and does not name the omission; Zika is the plausible candidate given the declared Flaviviridae gap, but that is inference. *Settled by:* the Investment Case portfolio tables read line by line against WHO's list.

Whether CEPI's two public statements of its own portfolio can be reconciled. CEPI's live priority pathogens page names eight pathogens plus Disease X and still navigates under "CEPI 2.0 and the 100 Days Mission", with no mention of CEPI 3.0; the Investment Case Annex 2 names a different and family-organised set, and omits Hantaviridae from its table while naming it in the surrounding prose. Both are current CEPI publications. Every family-level CEPI cell in section 3 rests on the Investment Case rather than on the live page, and would change if CEPI updated the page to something else. *Settled by:* CEPI updating its public list, or a statement of which document governs.

Whether HERA's Figure 1 cross-walk is reliable at cell level. Independent checking found that its CEPI column ticks Orthomyxoviridae and omits Poxviridae — both contradicted by CEPI's own publications — and that four of its thirteen columns are family-level ticks that render as if they attached to individual viruses. HERA itself disclaims the figure: it "may contain inaccuracies, omissions or inconsistencies". *Settled by:* cell-by-cell verification against each source list before any cell is quoted. The underlying cross-walk remains the single most useful artefact in this space, but it should be used as an index, not as evidence. Note that the figure's structure was itself a source of error in an earlier draft of this note, in both directions: its legend describes a merged bold convention while its column heads separate priority from prototype four ways. Both are true of different parts of the figure.

Whether any peer-reviewed cross-walk of these frameworks exists at pathogen level with priority and prototype separated. None was found in this survey. The only cross-walk located that separates the two designations is HERA's Figure 1, which is a staff working document that disclaims its own accuracy and which this note has shown to contain at least two wrong cells. Pulik et al. is a scoping review, not a cross-walk, and compares a different set of 13 resources that excludes UKHSA, CEPI, HERA and ECDC. The gap is therefore larger than this note previously claimed, not smaller: there is no peer-reviewed cross-walk at all, and the one non-peer-reviewed artefact that does the job is unreliable at cell level. *Settled by:* someone doing it — which, given the material above, is now a contained piece of work rather than a project.

Whether tools demonstrably transfer within a viral family. No empirical test was found in the peer-reviewed literature by the search described in section 4, which was informal. *Settled by:* either a properly specified systematic search, or a study that takes a prototype-derived asset — an assay, a reagent panel, an immunogen-design rule, a trained model — and measures its performance on held-out family members. This is the gap the project itself could partly fill, and saying so is a stronger framing than claiming the transfer assumption as background.

Whether NIAID's framework is still operative US policy. The December 2021 plan is still the posted document and the prototype list has not been revised since October 2023, but reporting from February 2026 indicates the NIAID 2026–2030 strategic plan omits pandemic preparedness and biodefense, that staff were instructed to remove such language from the website, and that the acting director departed in May 2026. This is secondary-sourced and should be flagged as

such. *Settled by:* the NIAID strategic plan itself, which was not retrievable at its usual URL. Related and unresolved: NIAID's Category A/B/C biodefense page returned HTTP 403 on 20 October 2026, so the contents of the third NIAID list could not be verified directly; it is cited as live by both HERA and Pulik et al.

Items closed by this revision

Three claims carried as uncertain in the previous draft are now verified and should not be flagged to a reader as open.

- **The PHEMCE financial figures.** $15.65bn for pandemic influenza and $645m for coronaviruses are both in the Multiyear Budget FY2025–29 Appendix C table (p.41), as $15,654m and $645m. The 23% share is correct and its denominator is now stated: the total estimated five-year need is $66,888m (Appendix C), given in the executive summary as "$66.9 billion" (p.1). 15,654 / 66,888 = 23.4%.
- **The Select Agents claims.** All three are confirmed against the HHS/CDC final rule at 89 FR 101941 (17 December 2024), effective 16 January 2025. §73.3(b) lists SARS-CoV at (19) and SARS-CoV/SARS-CoV-2 chimeric viruses at (20), with no entry for SARS-CoV-2 itself or for MERS-CoV; the 1918 entry at (16) reads exactly as quoted in section 2. The chimera entry at (20) is a nuance this note previously omitted and which a reader should have: SARS-CoV-2 is unregulated as itself but regulated as a component of a SARS-CoV chimera.
- **The CEPI 3.0 cluster.** The launch date, the 2027–31 scope, the viral-family shift, the "eight of the nine" commitment, the named strategic gaps and the ACE-2-binding sarbecovirus exclusion are sourced across three separate CEPI documents plus a live press release, not one PDF. The financial figures come from a fourth, the 2025 Annual Progress Report — where the US$214.5m coronavirus figure is itself contradicted by CEPI's own live page, which says US$230m across the same 14 projects. What remains open is that inconsistency and the larger one with CEPI's live priority-pathogens page, recorded as its own item above.